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Breaking Amides Using Nickel Catalysis

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ABSTRACT: Amides have been widely studied for decades, but their synthetic utility has remained limited in reactions that proceed with rupture of the amide C–N bond. Using Ni-catalysis, we have found that amides can now be strategically employed in several important transformations: esterification, transamidation, Suzuki–Miyaura couplings, and Negishi couplings. These methodologies provide exciting new tools to build C–heteroatom and C–C bonds using an unconventional reactant (i.e., the amide), which is ideally suited for use in multistep synthesis. It is expected that the area of amide C–N bond activation using non-precious metals will continue to flourish and, in turn, will promote the growing use of amides as synthons in organic synthesis.

KEYWORDS: nickel, catalysis, cross-coupling, amides, non-precious metal

1. Introduction

Transition metal-catalyzed cross-couplings have revolutionized the way chemists assemble small molecules.¹ Despite the long-standing prevalence of palladium catalysis in this field, much interest has recently been devoted to the use of base metal catalysis to enable coupling reactions (Figure 1).² Our laboratory³ and others have focused on the use of nickel catalysis due to: (a) nickel's natural abundance and consequential low cost (\$720/oz. for Pd vs \$0.47/oz for Ni);^{2d–h} (b) favorable toxicological profiles for orally administered drugs (permissible daily exposure: 100 µg/day Pd vs 500 µg/day Ni);⁴ (c) opportunities in green chemistry;^{3g–i,5} and (d) the potential to develop new chemical transformations based on alternate reactivity profiles.

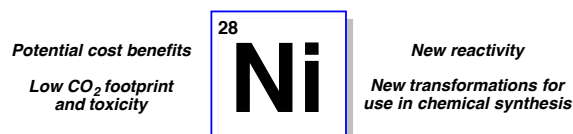


Figure 1. Benefits and opportunities of nickel catalysis.

Although generally less appreciated compared to palladium catalysis, it should not go unnoticed that nickel has been employed in industrial settings, in some cases for many decades.⁶ Several notable examples are depicted in Figure 2. The first features DuPont's hydrocyanation of butadiene (1). This transformation has been used to mass-produce adiponitrile (2), the precursor to Nylon 66, since 1971.^{6a} Another notable example is the Shell Higher Olefin Process (SHOP), where ethylene (3) can be converted to α -olefins 4 en route to detergents and plasticizers. This nickel-catalyzed process is used to

manufacture >1 million tonnes of olefins 4 per year.^{6b} Finally, the use of nickel catalysis in the pharmaceutical industry is gaining traction. One notable example involves the synthesis of Lyrica (5), wherein the primary amine is introduced via Raney Ni reduction of the corresponding nitrile on >2 tonne scale.^{6c} With regard to cross-coupling, the Genentech process team has shown that PI3K inhibitor 6 could be accessed using a nickel-catalyzed Suzuki–Miyaura coupling performed on >50 kg scale.^{6d} These large-scale examples bode well for future industrial applications of nickel-catalysis in chemical manufacturing processes.

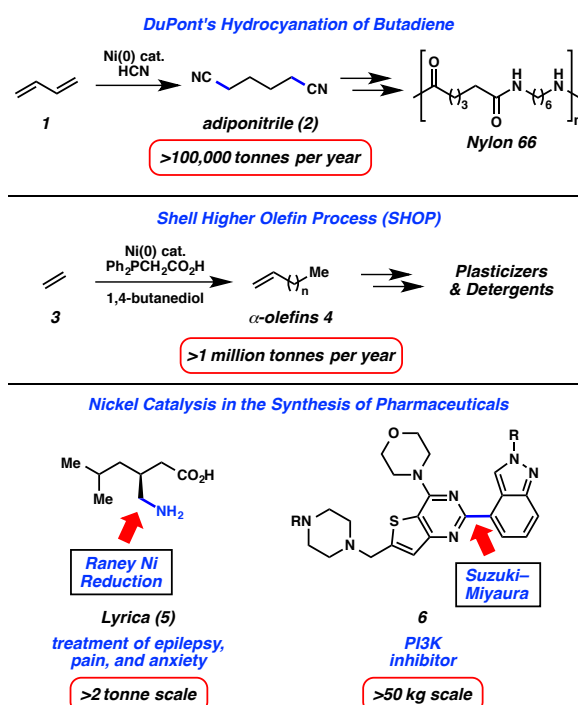


Figure 2. Industrial examples of nickel catalysis.

Given the potential benefits of nickel catalysis, it is not surprising that many academic and industrial groups have sought to develop new nickel-based methodologies (Figure 3). For example, intermolecular C–C bond forming reactions using unconventional substrates such as carbonates, carbamates, esters, and ethers have been reported for both aryl^{2e,3a–i,7} and benzylic⁸ electrophiles. Nickel has also been strategically employed in Heck reactions,⁹ aminations,¹⁰ and reductive couplings.¹¹ Other notable examples involving nickel catalysis include its synergistic use in photoredox catalysis,¹² in addition to efforts in C–H¹³ and C–F bond activation.¹⁴ Finally, the activation of amide C–N bonds^{3j–o,15} using nickel catalysis, which is the topic of this Perspective review, has recently emerged as a powerful synthetic tool.

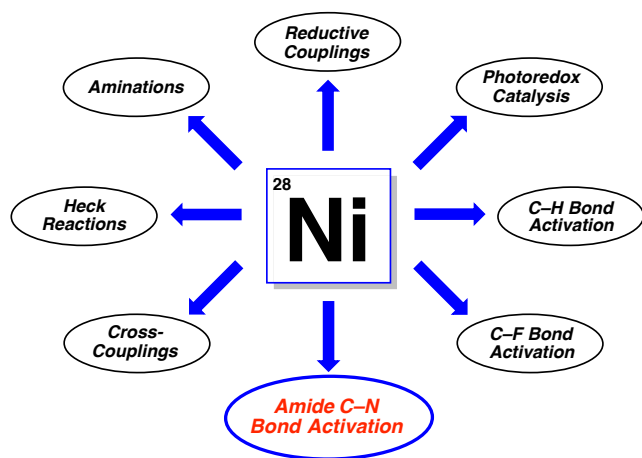


Figure 3. Select examples of recent advances in nickel catalysis, including the nickel-catalyzed activation of amides.

The efficient cleavage of amide C–N bonds remained an underdeveloped process for many decades.¹⁶ Such reactivity is challenging due to the well-known resonance stabilization of amides as postulated by Pauling in the 1950s (Figure 4).^{16b} In fact, the repertoire of reactions involving amide C–N bond cleavage that are generally considered useful to synthetic chemists are limited. Notable transformations include: (a) the conversion of Weinreb amides to ketones using basic and pyrophoric reagents (**7**→**8**),¹⁷ (b) the stoichiometric reduction of amides using Schwartz's zirconium-based reagent (**9**→**10**),¹⁸ and (c) the hydrolysis or esterification of amides (**9**→**11**), which often requires harshly acidic or basic reaction conditions, a large excess of the nucleophile, high temperatures, or a combination of these reaction parameters.^{16a}

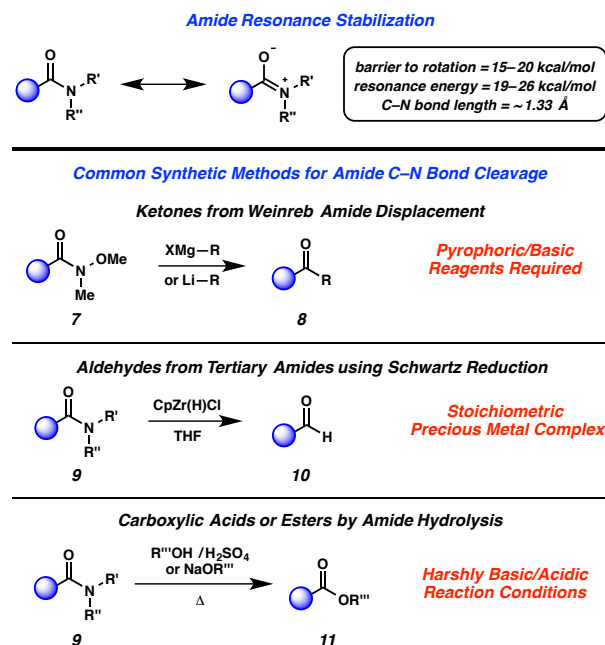


Figure 4. Amide stability and common synthetic methods involving amide C–N bond cleavage.

We reasoned that amides, despite being underutilized in amide C–N bond cleavage reactions, presented a unique opportunity. A new strategy to break amide C–N bonds could overcome a classical problem in organic chemistry, while also providing a new synthetic tool. For example, as suggested in Figure 5, we view amides as being ideally suited for use in multistep synthesis. They are well known to be stable to an array of reaction conditions (e.g., acids, bases, redox, Pd catalysis, heat, etc.) and can therefore be carried through multistep sequences before, in principle, carrying out a late-stage transformation of the amide. As an additional feature, amides can be exploited as directing groups in C–H functionalization reactions.¹⁹ Classically, this has been achieved by ortho-metallation of arene substrates, as exemplified by the lithiation of **12** to give reactive intermediate **13**.²⁰ Transition metal-catalyzed approaches to C–H functionalization of arenes and aliphatic systems have also been developed, as exemplified by Sanford's conversion of amide **14** to phenylated product **15**.²¹

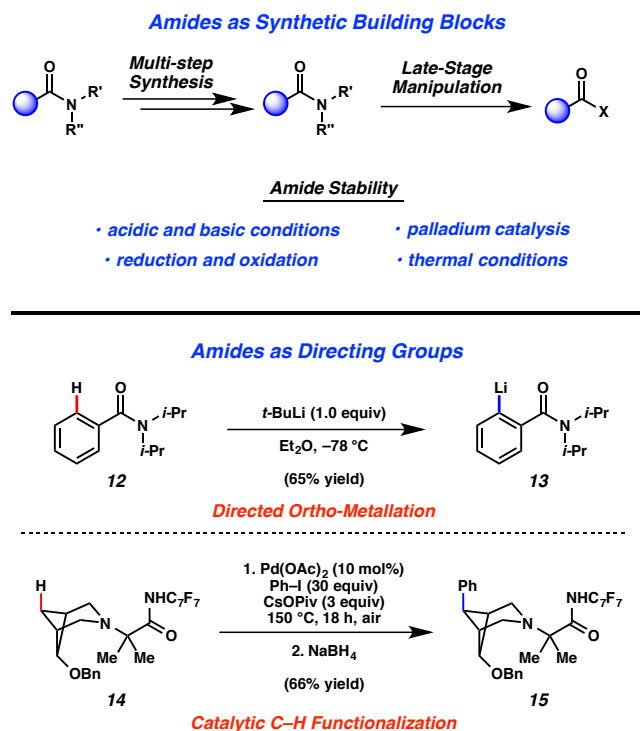


Figure 5. Opportunities for amides as synthetic building blocks.

We envisioned carrying out the sequence shown in Figure 6, wherein amides **9** would undergo Ni-catalyzed amide C–N bond activation to give acyl nickel species **16**.²² In situ trapping of **16** with a nucleophile would furnish acyl derivatives **17**. Following this paradigm, our laboratory has recently developed several C–C and C–heteroatom bond-forming reactions of amides to give ketones, esters, and amide products.^{3j–m,o} This Perspective highlights such efforts. Although not our focus herein, many other related advances deserve high praise, such as the nickel-catalyzed decarbonylative borylation by Shi,^{15a} Szostak's nickel-catalyzed Negishi coupling of amides to access biaryls,^{15c} Szostak's decarbonylative and non-decarbonylative Pd-catalyzed reactions of twisted amides,^{15e,23} Zou's Pd-catalyzed Suzuki–Miyaura coupling of amide derivatives,²⁴ and Murakami's insertion of alkenes into the C–N bonds of β -lactams.²⁵

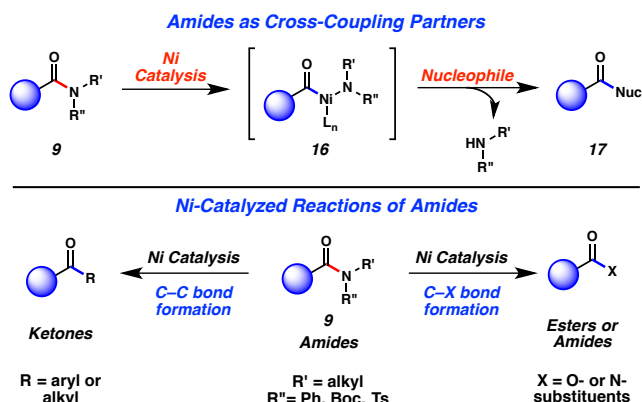


Figure 6. Nickel-catalyzed activation and reactions of amides studied in our laboratory.

2. Carbon–Heteroatom Bond Forming Reactions

Our first forays into the nickel-catalyzed manipulation of amides involved reactions with alcohols to give ester products (Figure 7).^{3j} The esterification of amides remained a challenging transformation for many decades, which rendered it an exciting starting point for our studies. Notable methods for the esterification of amides include classical alcoholysis under basic or acid conditions (often with heat and a large excess of nucleophile),^{16a} the nitrosation of *N*-methyl amides,²⁶ reactions of acyl aziridines,²⁷ or Keck's methylation/hydrolysis protocol.²⁸ The catalytic means to convert amides (**9**) to esters (**18**) using nickel catalysis suggested in Figure 7 had not previously been demonstrated.

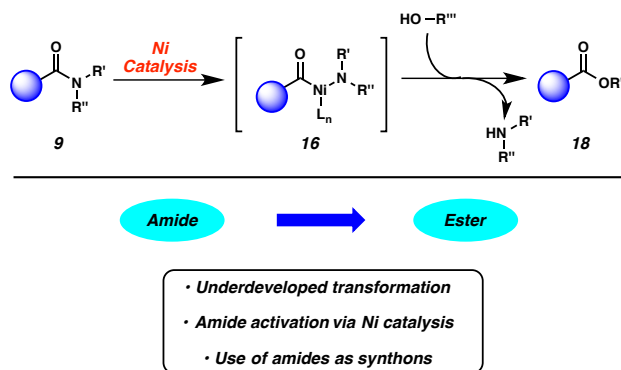


Figure 7. Design of the nickel-catalyzed esterification of amides.

An extensive survey of reaction parameters led to the identification of suitable reaction conditions for the amide to ester conversion. Specifically, it was found that an array of anilides **19** underwent the desired coupling with alcohols **20** in the presence of Ni(cod)₂ and the NHC ligand SIPr in toluene at 80 °C (Figure 8).^{3j} The transformation proceeds efficiently using only 1.2 equiv of the alcohol coupling partner. With regard to the amide component, benzamides possessing electron-withdrawing, electron-donating, and ortho-substituents were tolerated, as demonstrated by the formation of es-

ters **22–24**. Heterocyclic substrates, such as a furan and a quinoline, could also be used to generate esters **25** and **26**, respectively. With regard to the alcohol component, primary, secondary, and tertiary alcohols could all be utilized. Select examples include the use of (–)-menthol, 1-adamantol, and *N*-Boc prolinol, which gave **27–29**. Moreover, esters **30** and **31** were obtained when an indolyl alcohol and steroidal alcohol were used, respectively.

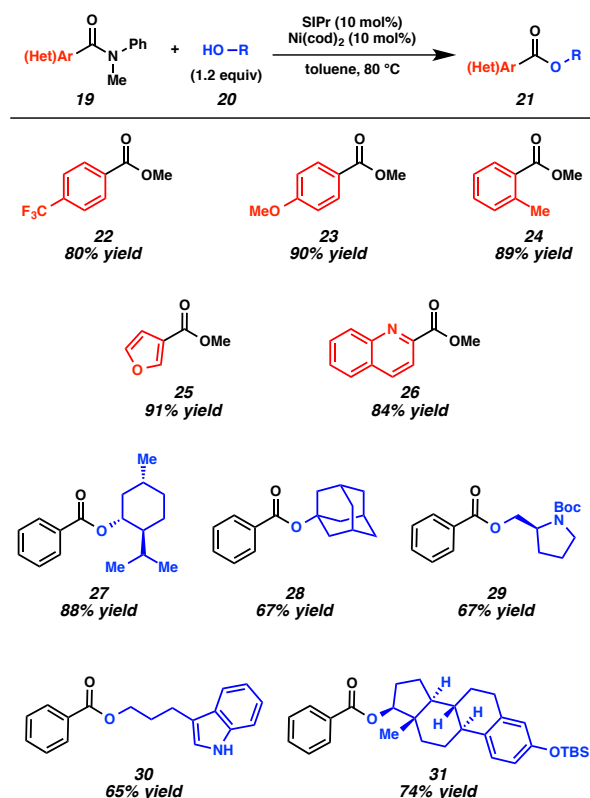


Figure 8. Selected examples of the nickel-catalyzed esterification of amides methodology.

The presumed catalytic cycle for the nickel-catalyzed esterification of amides is shown in Figure 9. The reaction is thought to proceed via oxidative addition (**32**+**33**→**34**), ligand exchange (**34**→**35**), and reductive elimination (**35**→**36**+**33**). Our collaborators, Professor Ken Houk and coworkers, performed calculations that were consistent with this pathway, along with insightful calculations regarding the overall thermodynamics of the transformation. Oxidative addition is thought to be the rate-determining step with a calculated barrier of 26.0 kcal/mol. The three-centered oxidative addition transition state obtained from the Houk group's calculations is shown. It should be noted that Shi's laboratory has obtained experimental evidence for the oxidative addition step.^{15a}

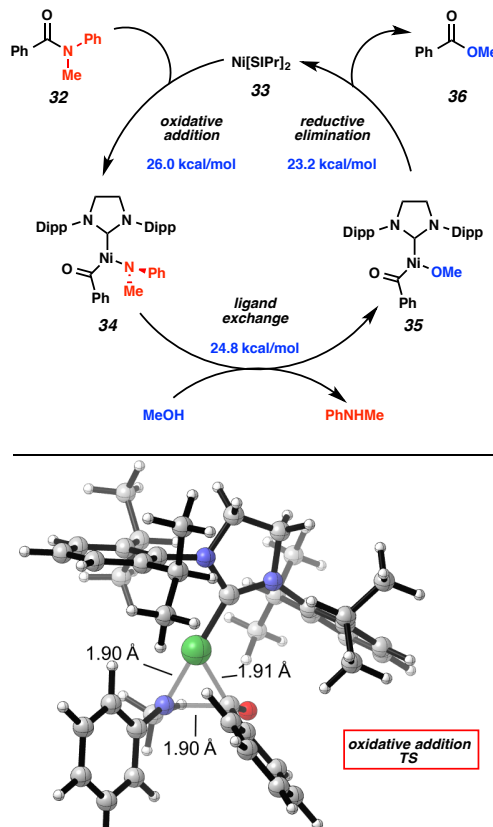


Figure 9. Proposed mechanism of the nickel-catalyzed esterification of amides.

We also examined a series of selectivity studies, one of which is highlighted in Figure 10. In this case, we prepared bis(amide) **37** and subjected it to nickel-catalyzed esterification using 1.2 equiv of (–)-menthol (**38**). This led to the selective esterification of the benzamide-derived anilide, without disruption of the proline-derived anilide. Additionally, the proline-based leaving group **39** in this transformation was recovered in 83% yield, without loss of enantiopurity. Thus, this result not only demonstrates that selective cleavage of polyamides is possible, but also highlights the mild nature of the reaction conditions.

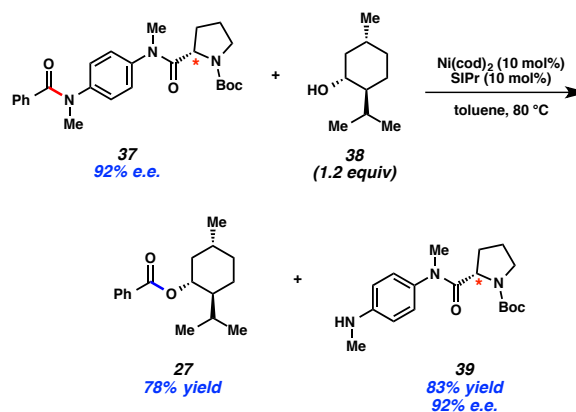


Figure 10. Selective activation of acyl amide C–N bond over alkyl amide C–N bond.

An additional example showcasing the mild and selective esterification of amides is provided in Figure 11. We prepared valine-derivative **40**, which possesses an amide, an ester, and an epimerizable stereocenter. Upon treatment of **40** with (–)-menthol (**38**, 1.2 equiv) under Ni-catalyzed reaction conditions, we obtained ester **27** and liberated enantioenriched amine **41**. Of note, the amide was selectively cleaved, while the ester remained intact.

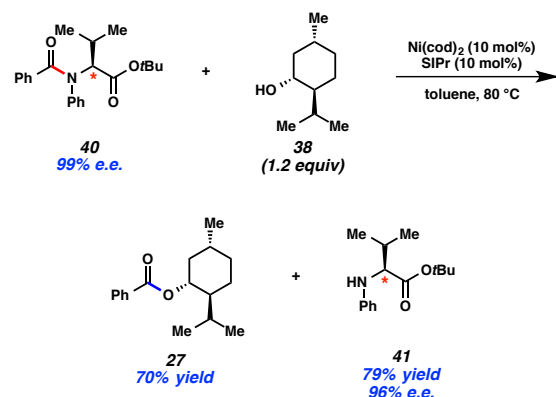


Figure 11. Ni-catalyzed esterification of valine-derived amide **40**, without disturbing an ester.

Having demonstrated that amides can be employed in nickel-catalyzed couplings, we sought to examine other nucleophiles in related processes. Amines were deemed an intriguing class of nucleophiles for reaction development, as their use would allow for the unusual transamidation reaction to be achieved.^{3m} Transamidation is a classic reaction that has seen relatively little success in the literature over the past few decades.²⁹ Transamidation of primary amides is the most well developed.³⁰ With regard to the transamidation of secondary amides, breakthroughs include earlier studies by Bertrand³¹ and Gellman and Stahl.³²

The challenges associated with the transamidation of secondary amides are summarized in Figure 12. Kinetically, the transformation is plagued by the unusually high activation barrier for amide C–N bond cleavage.¹⁶ Additionally, the overall reaction (**42**+**43**→**44**+**45**) is typically a thermoneutral process and may result in equilibrium mixtures, as has been seen in prior transamidation studies.³² With these considerations in mind, we proposed to achieve the net transamidation using a simple two-step sequence. Secondary amides **42** would first undergo *N*-functionalization with an electron-withdrawing group ('Z') to give amide derivative **46**, which would presumably be activated toward oxidative addition. In the second step, treatment of **46** with an amine nucleophile under catalytic nickel conditions would lead to oxidative addition intermediate **47**, which, in turn, would undergo trapping with amine **43** to yield amide **44**. The reaction would be driven thermodynamically by the release of amine **48**, bearing an elec-

tron-withdrawing group. From our experience with esterification chemistry, we had observed that *N*-Boc amides could undergo nickel-catalyzed activation, just like *N*-Ph substrates. Considering the ease by which a Boc group can be introduced onto a secondary amide, we opted to pursue *N*-Boc functionalization as our mode of activating the amide substrates.

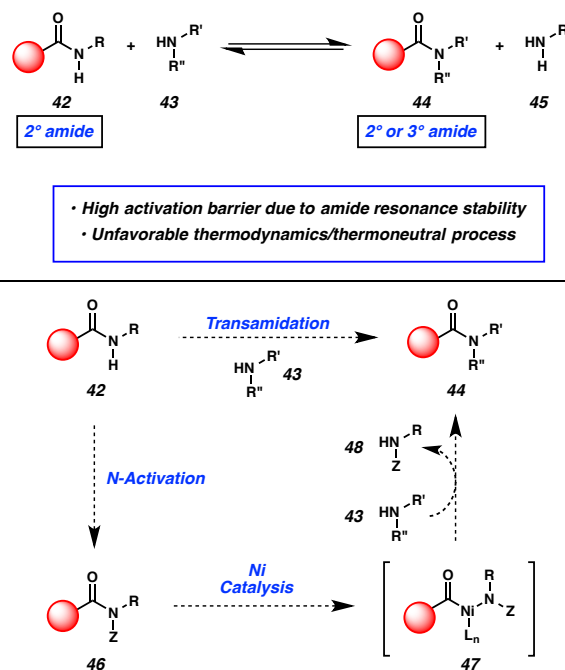


Figure 12. Challenges associated with the transamidation of secondary amides and our two-step approach.

As shown in Figure 13, we found that a range of Boc-activated secondary amides undergo nickel-catalyzed transamidation using secondary amine nucleophiles. With regard to the benzamide substrate, electron-withdrawing and electron-donating groups are tolerated, as shown by the formation of product **51**–**53**. Additionally, heterocycles could be employed to give products such as furan **54** and thiophene **55**. The scope with respect to the amine nucleophile was also found to be broad. Hindered amines could be used, as suggested by transamidation products **56** and **57**. Moreover, the formation of **58**–**60** suggests that the methodology tolerates heterocyclic amines, such as imidazolines, piperazines, pyridines, and carbazoles.

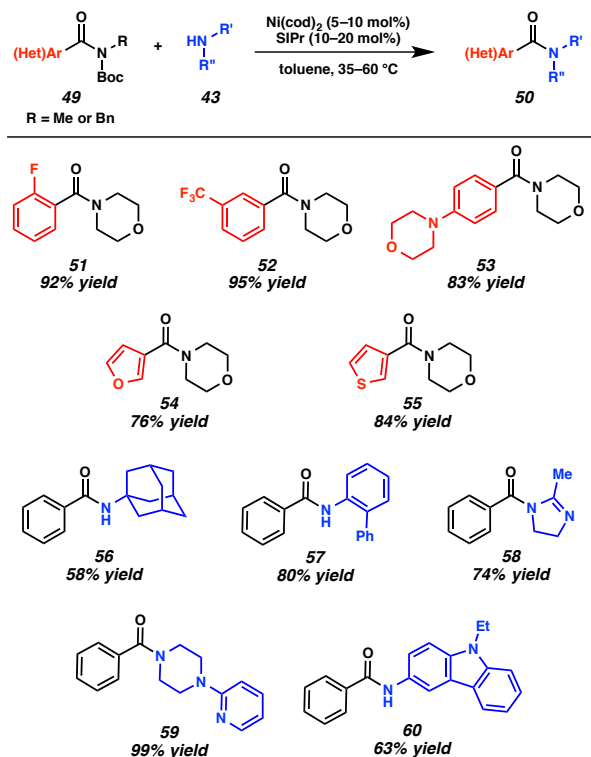


Figure 13. Selected examples of the nickel-catalyzed transamidation of Boc-activated amides.

The transamidation methodology is perhaps best illustrated by the results shown in Figure 14, where secondary amide **61** was converted to a series of amino acid-derivatives **62** using the two-step procedure. Boc-activation of **61** proceeded smoothly and in quantitative yield. Next, nickel-catalyzed transamidation using optically enriched amino esters delivered the desired amide products **64–69**. Derivatives of alanine, phenylalanine, valine, leucine, isoleucine, and proline could be utilized in this coupling methodology. In all cases, the ester withstood the reaction conditions and the transamidation reaction proceeded without loss of stereochemical integrity. This methodology provides one of the most general solutions to the classic problem of secondary amide transamidation reported to date.

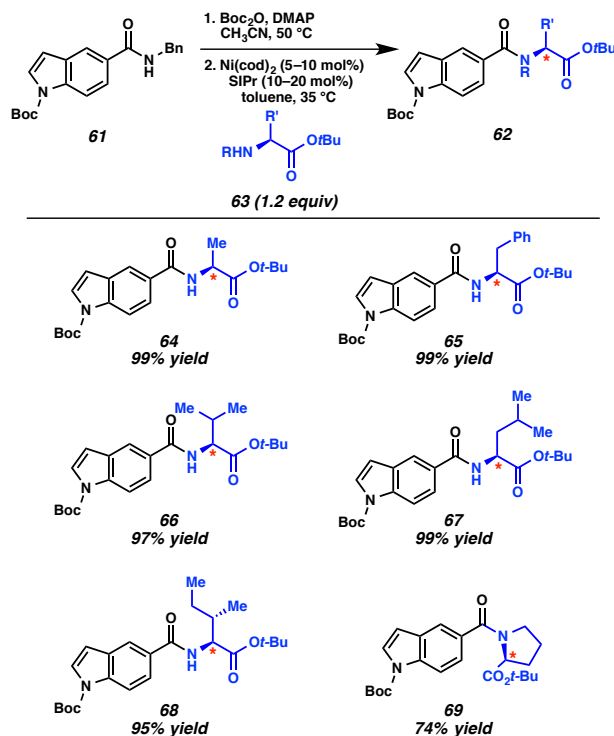


Figure 14. Scope of the two-step, nickel-catalyzed transamidation of amides using amino acid-derived nucleophiles.

3. Carbon–Carbon Bond Forming Reactions

Another exciting opportunity lay in the notion of using amides as building blocks for the construction of C–C bonds using catalysis. In contrast to popular Weinreb amide chemistry (Figure 15, **7**→**70**),¹⁷ such a method could potentially sidestep the use of pyrophoric, basic reagents and offer improved functional group compatibility. Thus, we set out to develop Ni-catalyzed cross-couplings of amides **46** that would allow for the synthesis of aryl and alkyl ketones **70**. We were also optimistic that these efforts, if successful, would lend some credence to the mechanistic notion described earlier (i.e., amides could undergo oxidative addition in the presence of an appropriate nickel catalyst, see Figure 9).

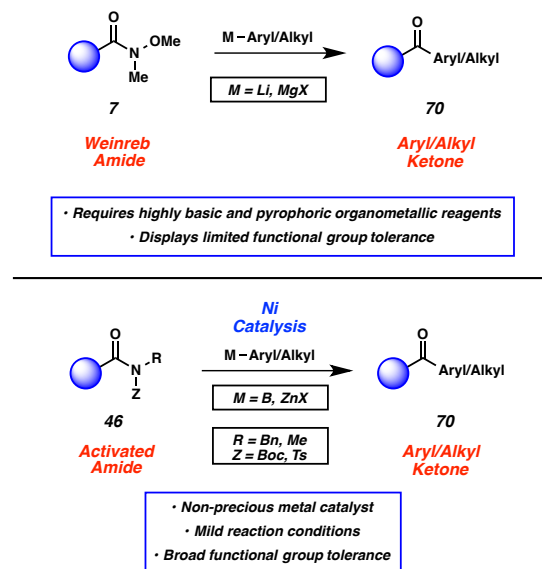


Figure 15. Comparison of Weinreb amide displacement chemistry and nickel-catalyzed cross-couplings of amides for C–C bond formation.

Of the possible C–C bond-forming cross-coupling reactions, we elected to initiate our studies by pursuing the Suzuki–Miyaura³³ coupling of amides. The Suzuki–Miyaura cross-coupling, typically performed on aryl halides or pseudohalides,^{34,35} has transformed the landscape of how chemists construct small molecules, especially molecules of medicinal importance.

After performing optimization studies, we were delighted to find that pinacolboronate esters (**71**) could serve as amide (**49**) cross-coupling partners to furnish ketone products **72** (Figure 16).^{3k} The methodology was found to be tolerant of substitution at the ortho, meta, and para positions, as shown by the formation of products **73a–c**. Additionally, substrates bearing either electron-donating or electron-withdrawing groups could be utilized, to give products **74**, **75**, and **76**. From the latter two of these examples, it should be emphasized that esters and ketones withstand the reaction conditions, which is typically not the case using Weinreb amide chemistry.¹⁷ Heterocycles could also be used, as shown by the formation of furan and thiophene products **77a** and **77b**, respectively. With regard to the scope of the boronic ester component, aryl nucleophiles could be utilized, including those with ortho-, meta-, or para-functionalities (see products **78a–c**). Moreover, a series of heterocyclic boronic esters, including indoles, pyrroles, pyrazoles, and furans, underwent the cross-coupling reaction to give ketones **79–82**, respectively. Finally, a robustness screen revealed that epoxides, tertiary alcohols, nitriles, secondary amides, and the free NHs of indoles could all be tolerated in this methodology.^{3k}

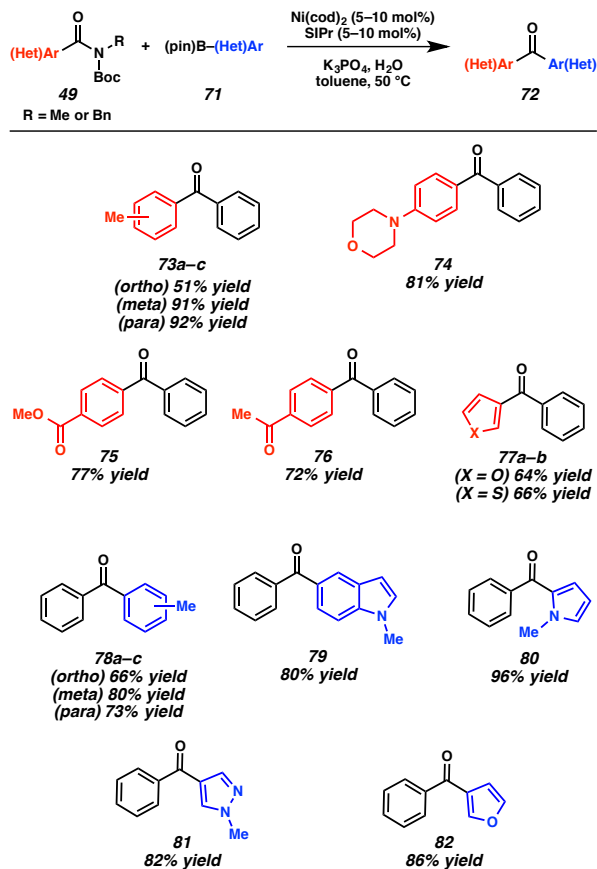


Figure 16. Selected examples of the nickel-catalyzed Suzuki–Miyaura coupling of amides.

We also pursued a series of experiments involving the Ni-catalyzed Suzuki–Miyaura cross-coupling of amides, with the hope of using this chemistry in tandem with other cross-coupling processes. The first, shown in Figure 17, shows the use of the two sequential nickel-catalyzed reactions of amides: the esterification reaction^{3j} and the Suzuki–Miyaura coupling.^{3k} Bis(amide) **83**, where one of the amides bears a Boc-activating group, was treated with (–)-menthol (**38**) under our Ni-catalyzed esterification conditions. This led to the formation of ester **84** in 90% yield. Of note, the secondary amide was not disturbed in this process. From intermediate **84**, we performed a straightforward Boc-activation of the secondary amide, which set the stage for the Ni-catalyzed Suzuki–Miyaura coupling. Using heterocyclic boronic ester **85**, we obtained ketone product **86** in 82% yield over two steps. The ester was not disturbed in this process. A key point to take-home from this sequence is that the selectivity in the initial coupling of **83** is strictly dependent on which amide possesses the Boc-activating group. As such, we have also shown the order of cross-couplings can be reversed (i.e., Suzuki–Miyaura coupling can take place prior to the esterification in the overall conversion of **83** to **86**).^{3k}

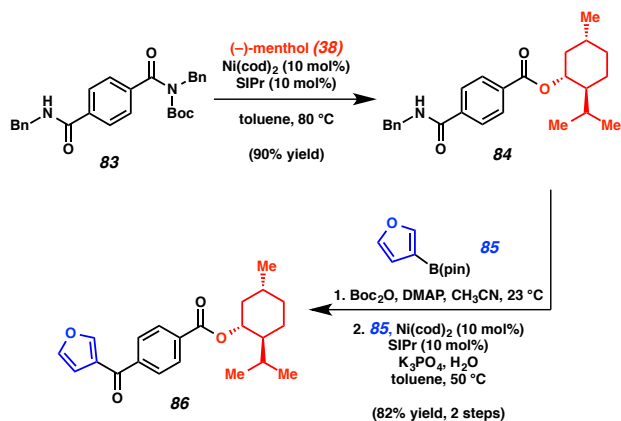


Figure 17. Sequential nickel-catalyzed esterification and Suzuki–Miyaura coupling of amides.

We were also eager to execute sequential Pd- and Ni-catalyzed Suzuki–Miyaura couplings (Figure 18). Boc-activated amide **87** was readily synthesized from the corresponding carboxylic acid. With this bi-functional substrate in hand, we first performed a Pd-catalyzed Suzuki–Miyaura coupling³⁶ with furanyl boronic acid **88** to deliver biaryl product **89**. The amide was not disturbed under these Pd-catalyzed, basic reaction conditions. From amide **89**, a Ni-catalyzed Suzuki–Miyaura coupling using pyrazole boronic ester **90** furnished **91**. This succession of cross-couplings demonstrates how our methodology can be used in strategic combination with traditional couplings to efficiently unite heterocyclic fragments.

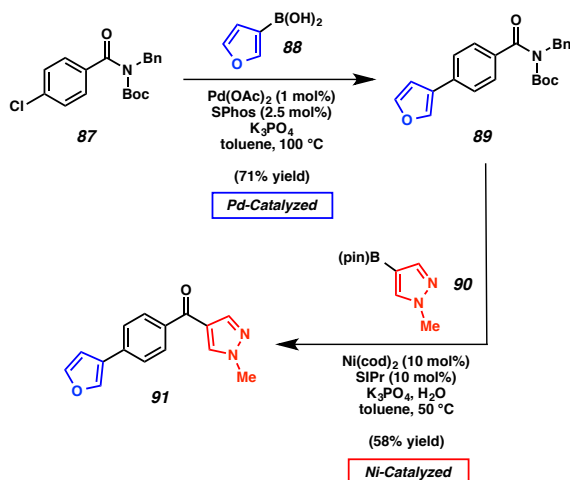


Figure 18. Sequential palladium- and nickel-catalyzed Suzuki–Miyaura couplings.

Having developed a means to access aryl–aryl ketones from amides, complementary to elegant methods discovered by Szostak^{15e,37,38} we sought to develop a corresponding protocol to assemble aryl–alkyl ketones. Unfortunately, the Suzuki–Miyaura coupling of aliphatic amide substrates or aliphatic boronates using the conditions described above were not successful. Thus, we

turned to the use of an alternate reaction, namely the Negishi coupling.³⁹ Ni-catalyzed Negishi couplings had previously been achieved using acyl halides,⁴⁰ anhydrides,⁴⁰ and thioesters,^{40,41} but never using amides.^{15c}

As shown in Figure 19, we found that the identity of the amide nitrogen substituents had a pronounced effect on the success of the Ni-catalyzed Negishi coupling of amides. For example, treatment of anilide **92a** (an excellent substrate for the Ni-catalyzed esterification) and benzylzinc bromide (**93**) with Ni(cod)₂ and SIPr in THF led to the recovery of anilide **92a** (entry 1). Alternatively, attempts to utilize Boc-activated secondary amide **92b**, a viable Suzuki–Miyaura coupling substrate, gave the desired ketone **94** in 60% yield (entry 2). Finally, we surveyed *N*-tosyl substrate **92c**, which underwent conversion to ketone **94** in 81% yield (entry 3). We elected to further pursue the cross-coupling of *N*-tosyl amide substrates when evaluating the scope of this transformation.

Entry	<i>N</i> -Substituent	Recovered 92	Yield of Ketone 94
1	<i>N</i> -Me, <i>N</i> -Ph (92a)	100%	0%
2	<i>N</i> -Boc (92b)	40%	60%
3	<i>N</i> -Ts (92c)	17%	81%

Figure 19. Amide *N*-substituent survey data for the nickel-catalyzed alkylation of amides.

A brief selection of examples that highlight the scope of this methodology, particularly with respect to the organozinc coupling partner, is shown in Figure 20. Non-branched nucleophiles could be used, as shown by the formation of ketone **98** in 80% yield. The syntheses of ketones **99–102** demonstrate that α -branched nucleophiles could also be employed to give products of sp^2 – sp^3 cross-coupling. Moreover, β -branching is tolerated, as shown by the coupling of a neopentyl nucleophile to give ketone **103**. It should also be noted that the methodology is tolerant of substitution on the arene (e.g., –NMe₂, –CF₃, –F, and –OMe).³¹

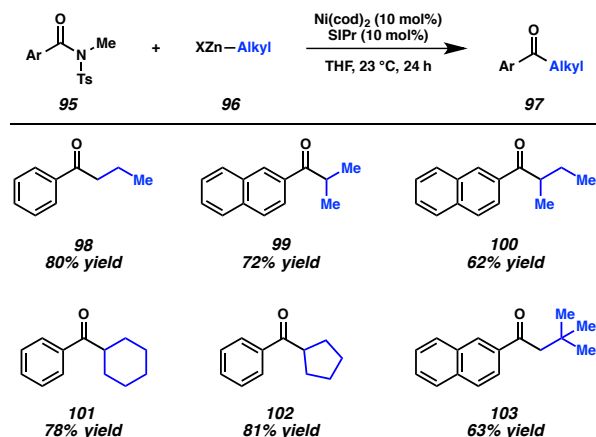


Figure 20. Scope of the organozinc halide coupling partner for the nickel-catalyzed alkylation of amides.

The utility of the Ni-catalyzed Negishi coupling of amides can be seen in the example provided in Figure 21. Substrate **104**, bearing an amide and a methyl ester, was coupled with cyclohexyl zinc iodide (**105**) under our typical reaction conditions. This reaction led to the formation of ketone **106** in 71% using 1 g of substrate. Of note, the ester was not disturbed under the reaction conditions. **106** is a precursor to Pfizer's glucagon receptor modulator **107**.⁴² Our synthesis of **106** compares favorably to the literature protocol, which had proceeded in 34% yield using Weinreb amide displacement chemistry.⁴²

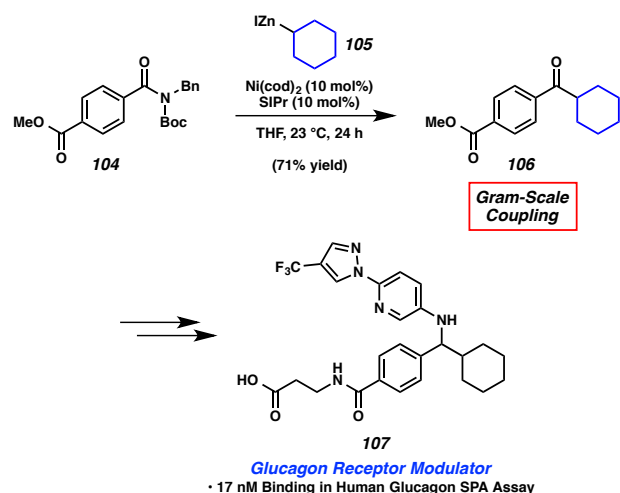


Figure 21. Application of the Ni-catalyzed alkylation of amides to the gram-scale synthesis of **106**.

4. From the Glovebox to the Benchtop

The methods described thus far provide exciting new tools for the manipulation of amides that allow for the assembly of C–O, C–N, and C–C bonds. However, one limitation of this methodology is that Ni(cod)₂, the precursor used throughout our studies, typically requires glovebox handling. Similarly, the free NHC ligand, SIPr, is unstable to benchtop conditions. Given this limi-

tation of our amide coupling methodology and the general utility of Ni(cod)₂ in unrelated catalytic reactions,⁴³ we sought to develop a means to handle Ni(cod)₂ on the benchtop.

Our approach³ⁿ is inspired by and mimics a recent methodology reported by Buchwald, wherein catalysts and ligands could be encapsulated in paraffin wax capsules to enable benchtop delivery.^{44,45} After developing a robust means to prepare paraffin capsules, we found that Ni(cod)₂ or Ni(cod)₂/SIPr mixtures could be readily encapsulated and used to promote reactions on the bench. Figure 22 highlights the Ni-catalyzed esterification (**108**+**38**→**27**) and transamidation (**109**+**110**→**111**) reactions using the paraffin capsules. Both reactions proceeded smoothly on the benchtop and with yields comparable to those reported using the typical glovebox conditions. Similar success was seen in our couplings of amides to form C–C bonds. For example, the Suzuki–Miyaura coupling of **112** and **113** delivered ketone **114** in 82% yield, whereas the Negishi coupling of **104** and **105** furnished ketoester **106** in 55% yield.

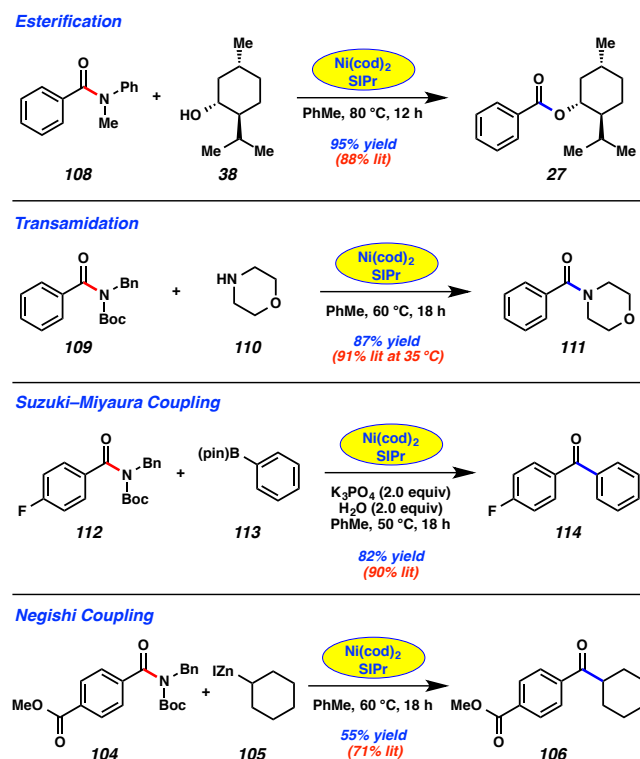


Figure 22. Paraffin encapsulation of Ni(cod)₂ and SIPr allow amide couplings to take place on the benchtop.

Beyond the examples shown in Figure 22, several features of the benchtop reactions should be noted. A) The capsules can be stored open to air, without appreciable loss of catalytic activity. For example, the esterification to access **27** proceeded in 95% yield using a Ni(cod)₂/SIPr-paraffin capsule that had been stored on

the benchtop for two months.³ⁿ B) The same transformation was performed on gram-scale (using a larger capsule), which delivered the ester product **27** in 97% yield.³ⁿ C) The Ni(cod)₂ capsules could be used to enable a host of other Ni-catalyzed reactions, such as Dong's oxidative esterification and amidation methodologies,⁴⁶ our laboratories aryl C–N bond forming reactions,^{3c} and Fu's sp²–sp³ cross-coupling of alkyl halides.⁴⁷ D) Paraffin capsules containing only Ni(cod)₂ or Ni(cod)₂ / SIPr combinations are being commercialized, which we expect will enable their more widespread use.⁴⁸

5. Summary

Amides have been widely studied for decades, but their synthetic utility has remained limited in reactions that proceed with rupture of the amide C–N bond. Using Ni-catalysis, we have found that amides can now be strategically employed in several new reactions: esterification,^{3j} transamidation,^{3m} Suzuki–Miyaura,^{3k} and Negishi couplings.^{3l} These methodologies, and those related breakthroughs described by other key players in the field,^{15,23,24,25} provide efficient new tools to build C–heteroatom and C–C bonds using an unconventional reactant (i.e., the amide), which is ideally suited for use in multistep synthesis.

The chemistry described herein lays the foundation for many opportunities when it comes to future development. Can the scope of the amide *N*-substituents be improved to encompass dialkyl amides, non-activated secondary amides, or primary amides? Can conditions be developed to enable the coupling of amides derived from aliphatic precursors, rather than aryl and hetaryl substrates? Can other nucleophiles or trapping reagents be used? Can computations, kinetic studies, and other experiments be used to uncover mechanistic details? Can the methods discussed herein be utilized to construct molecules of importance, such as drug candidates or natural products?⁴⁹ Finally, now that amides can be activated using catalysis, are there new transformations involving amides that remain to be discovered? Although only time will tell, recent studies by our laboratory^{3o} and others¹⁵ suggest the answer to many of these questions is 'yes'.

It is our hope and expectation that the area of amide C–N bond activation using non-precious metals will continue to flourish and, in turn, will promote the growing use of amides as synthons in organic synthesis.

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- ⁴⁵ For the use of paraffin to allow for the handling of other air-sensitive chemicals, see: (a) Taber, D. F.; Li, H.-Y. U.S. Patent 20050288257, Dec. 29, 2005. (b) Fang, Y.; Liu, Y.; Ke, Y.; Guo, C.; Zhu, N.; Mi, X.; Ma, Z.; Hu, Y. *Appl. Catal., A* **2002**, *235*, 33–38. (c) Taber, D. F.; Frankowski, K. J. *J. Org. Chem.* **2003**, *68*, 6047–6048. (d) Taber, D. F.; Nelson, C. G. *J. Org. Chem.* **2006**, *71*, 8973–8974. For examples of reagents encapsulated in paraffin, see: (e) Kosak, K. M.; Kosak, M. K. U.S. Patent 5413924, May 9, 1995. (f) Shen, C.; Spannenberg, A.; Wu, X.-F. *Angew. Chem., Int. Ed.* **2016**, *55*, 5067–5070.
- ⁴⁶ Whittaker, A. M.; Dong, V. M. *Angew. Chem., Int. Ed.* **2015**, *54*, 1312–1315.
- ⁴⁷ Zhou, J.; Fu, G. C. *J. Am. Chem. Soc.* **2004**, *126*, 1340–1341.
- ⁴⁸ Paraffin capsules containing Ni(cod)₂ and Ni(cod)₂/SIPr will soon be available commercially from TCI (TCIchemicals.com) and Sigma-Aldrich. The capsules and their TCI product numbers are as follows (Sigma-Aldrich product numbers are pending): Bis(1,5-cyclooctadiene)nickel(0) (Wax encapsulated): B5417. Bis(1,5-cyclooctadiene)nickel(0)–1,3-Bis(2,6-diisopropylphenyl)imidazolidin-2-ylidene (Wax encapsulated): B5418.
- ⁴⁹ For the prevalence of amides in pharmaceuticals, see: (a) Ghose, A. K.; Viswanadhan, V. N.; Wendoloski, J. J. *J. Comb. Chem.* **1999**, *1*, 55–68. (b) Brown, D. G.; Boström, J. *J. Med. Chem.* **2016**, *59*, 4443–4458.

SYNOPSIS TOC

